$x_1, \dots, x_N$ in V defines an isomorphism of TV with the free algebra $k\langle x_1, \dots, x_n \rangle$.

Also, one can make the following definition.

Definition 2.12.1. (i) The **symmetric algebra** SV of V is the quotient of TV by the ideal generated by $v \otimes w - w \otimes v$, $v, w \in V$.

(ii) The **exterior algebra** $\wedge V$ of V is the quotient of TV by the ideal generated by $v \otimes v$, $v \in V$.

(iii) If V is a Lie algebra, the **universal enveloping algebra** $\mathcal{U}(V)$ of V is the quotient of TV by the ideal generated by $v \otimes w - w \otimes v - [v, w]$, $v, w \in V$.

It is easy to see that a choice of a basis $x_1, \dots, x_N$ in V identifies SV with the polynomial algebra $k[x_1, \dots, x_N]$, $\wedge V$ with the exterior algebra $\wedge_k(x_1, \dots, x_N)$, and the universal enveloping algebra $\mathcal{U}(V)$ with one defined previously.

Moreover, it is easy to see that we have decompositions

$$SV = \bigoplus_{n \geq 0} S^n V, \ \wedge V = \bigoplus_{n \geq 0} \wedge^n V.$$

2.13. Hilbert's third problem

Problem 2.13.1. It is known that if A and B are two polygons of the same area, then A can be cut by finitely many straight cuts into pieces from which one can make B (check it — it is fun!). David Hilbert asked in 1900 whether it is true for polyhedra in three dimensions. In particular, is it true for a cube and a regular tetrahedron of the same volume?

The answer is "no", as was found by Dehn in 1901. The proof is very beautiful. Namely, to any polyhedron A, let us attach its "Dehn invariant" $D(A)$ in $V = \mathbb{R} \otimes (\mathbb{R}/\mathbb{Q})$ (the tensor product of $\mathbb{Q}$-vector spaces). Namely,

$$D(A) = \sum_a l(a) \otimes \frac{\beta(a)}{\pi},$$

where a runs over edges of A and $l(a), \beta(a)$ are the length of a and the dihedral angle at a.